Introduction to APIs and JSONs

INTERMEDIATE IMPORTING DATA IN PYTHON



Hugo Bowne-Anderson
Data Scientist at DataCamp

datacamp

APIs

- Application Programming Interface
- Protocols and routines
  - Building and interacting with software applications



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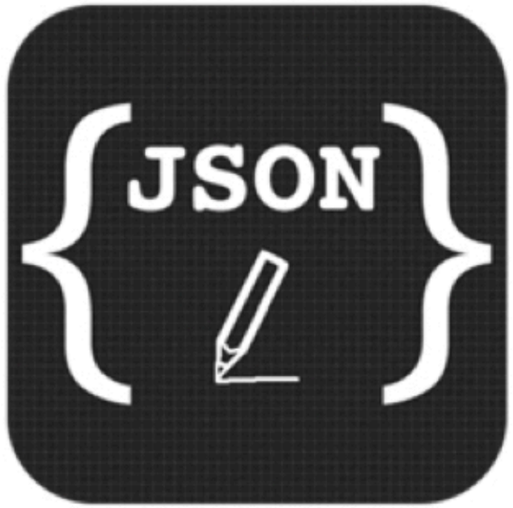
OMDb API

The Open Movie Database



JSONs

- JavaScript Object Notation
- Real-time server-to-browser communication
- Douglas Crockford
- Human readable



JSONs

```
{'Actors': 'Samuel L. Jackson, Julianna Margulies, Nathan Phillips,
 Rachel Blanchard',
 'Awards': '3 wins & 7 nominations.',
 'Country': 'Germany, USA, Canada',
 'Director': 'David R. Ellis',
 'Genre': 'Action, Adventure, Crime',
 'Language': 'English',
 'Rated': 'R',
 'Released': '18 Aug 2006',
 'Runtime': '105 min',
 'Title': 'Snakes on a Plane',
 'Type': 'movie',
 'Writer': 'John Heffernan (screenplay), Sebastian Gutierrez (screenplay),
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 'imdbID': 'tt0417148',
 'imdbRating': '5.6',
 'imdbVotes': '114,668'}
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Loading JSONs in Python

```
import json
with open('snakes.json', 'r') as json_file:
        json_data = json.load(json_file)
```

```
type(json_data)
```

```
dict
```

Exploring JSONs in Python

```
for key, value in json_data.items():
    print(key + ':', value)
```

```
Title: Snakes on a Plane
Country: Germany, USA, Canada
Response: True
Language: English
Awards: 3 wins & 7 nominations.
Year: 2006
Actors: Samuel L. Jackson, Julianna Margulies
Runtime: 105 min
Genre: Action, Adventure, Crime
imdbID: tt0417148
Director: David R. Ellis
imdbRating: 5.6
Rated: R
Released: 18 Aug 2006
```

Let's practice!

INTERMEDIATE IMPORTING DATA IN PYTHON

APIs and interacting with the world wide web

INTERMEDIATE IMPORTING DATA IN PYTHON



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Herein, you'll learn

- What APIs are
- Why APIs are important
- In the exercises:
  - Connecting to APIs
  - Pulling data from APIs
  - Parsing data from APIs

What is an API?

- Set of protocols and routines
- Bunch of code
  - Allows two software programs to communicate with each other



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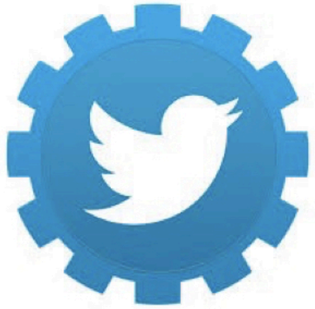




APIs are everywhere

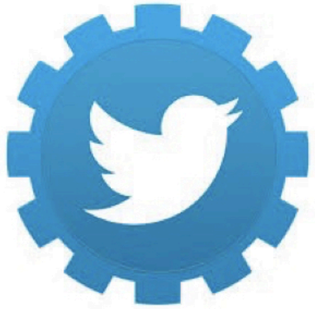


APIs are everywhere





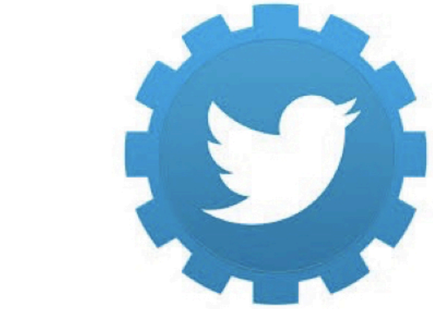
APIs are everywhere



UBER Developers



APIs are everywhere



UBER Developers





Instagram API

Connecting to an API in Python

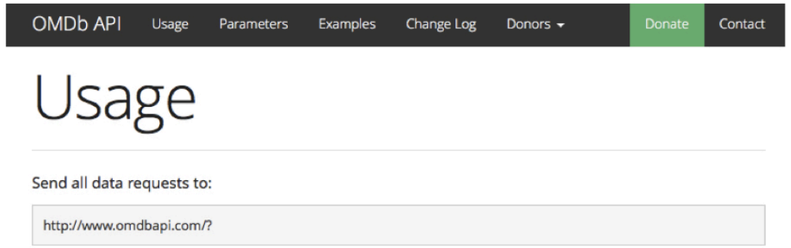
```
import requests
url = 'http://www.omdbapi.com/?t=hackers'
r = requests.get(url)
json_data = r.json()
for key, value in json_data.items():
    print(key + ':', value)
```

What was that URL?

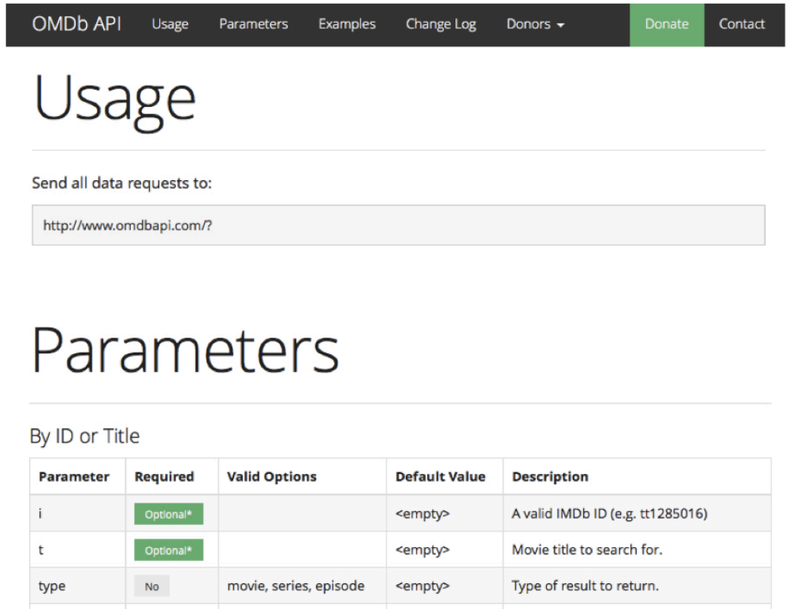
- http - making an HTTP request
- **www.omdbapi.com** - querying the OMDB API
- `?t=hackers`
  - Query string
  - Return data for a movie with title (t) ‘Hackers’

```
'http://www.omdbapi.com/?t=hackers'
```

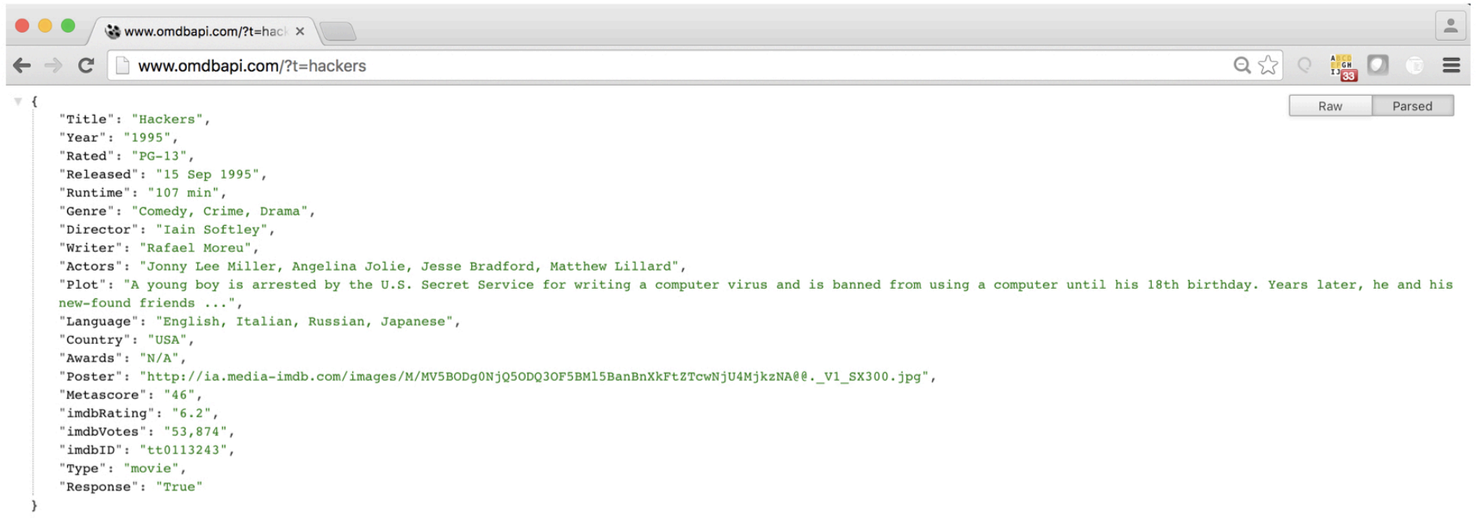
OMDb API


OMDb API
Usage
Parameters
Examples
Change Log
Donors
Donate
Contact
Usage
Send all data requests to:
http://www.omdbapi.com/?

OMDb API


OMDb API
Usage
Parameters
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Usage
Send all data requests to:
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Parameters
By ID or Title
Parameter
Required
Valid Options
Default Value
Description
i
Optional*
<empty>
A valid IMDb ID (e.g. tt1285016)
t
Optional*
<empty>
Movie title to search for.
type
No
movie, series, episode
<empty>
Type of result to return.

It's a regular URL!



Let's practice!

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